interphAIse

Can an LLM choose the next useful materials calculation?

From a controlled search benchmark to a prospective three-coating PET-MAD comparison

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THE RESULT

Feedback helps. The advantage is conditional.

20/30

Sulfide successes with feedback

Sequential 2/30, shuffled 6/30, random 8/30.

12 vs 24

On LLZO, random beats iterative

Feedback helps a narrow prior, but too late in ten queries.

2 held
2 missed

Four prospective adhesion forecasts

$LiTaF_6$ matched both ranges. $LiCaAlF_6$ fell below both.

The LLM replaces part of the selection judgment

The question is what to calculate next, and what result to expect.

Human task being delegated	LLM contribution	Grounding / control
Propose a promising direction	Select a coating and explain why	Frozen candidate IDs and budget
Learn from a failed test	Relate outcomes to chemistry	Only observed feedback is exposed
Decide the next calculation	Revise priorities after feedback	No-feedback and shuffled controls
Anticipate the result	Forecast adhesion and weak contact	PET-MAD tests the prediction

The LLM supplies the scientific decision policy. The tools supply the numerical evidence.

ORIGINAL INSPIRATION

From mechanism proposals to material choices

Both delegate a scientific choice, then test it with calculations.

	HELLMM	interphAIse
What the LLM imagines	Possible intermediates and mechanisms	Promising materials and expected responses
What the tools test	Structures, pathways and energetics	Compatibility and contact adhesion
Where intuition is replaced	Which mechanisms should exist in the study?	Which material deserves the next test?
Distinct contribution here	Richer chemical hypothesis generation	Controlled evidence on feedback use

Narrower creativity than HELLMM. A clearer experiment on sequential decision-making.

Two days produced two complementary experiments

The cheap study measures selection. The physical pilot tests forecasts.

Stage	Scope	What it can establish
21 Oct: frozen benchmark	50 structures, 43 formulas 150 runs across five arms	Whether feedback helps on one fixed task
21 Oct: environment study	141 structures, 100 formulas 3 environments, 450 runs	Where the decision policy transfers or fails
22 Oct: physical pilot	3 coatings, 2 contacts each 26 new jobs, 2 LLM choices	Prospective predictions and model sensitivity

These are separate evidence layers. The larger pool includes the original 50 structures.

“Acceptable” means passing a frozen proxy

$LiCoO_2$ at 4.0 V, paired with each electrolyte. Lower penalty is better.

P = max(self-decomposition, cathode contact, electrolyte contact)

Acceptance	Ranking resolution	Meaning
P ≤ 0.050 eV / non-Li atom Band gap ≥ 2 eV	0.020 eV / non-Li atom bins	Additional reaction on contact, relative to equilibrated endpoints

Lithium transport, finite-temperature behavior, synthesis and cycling were not validated.

The exhaustive table is cheap. Search on it is a controlled benchmark, not a compute-saving result.

Five arms separate prior knowledge from feedback

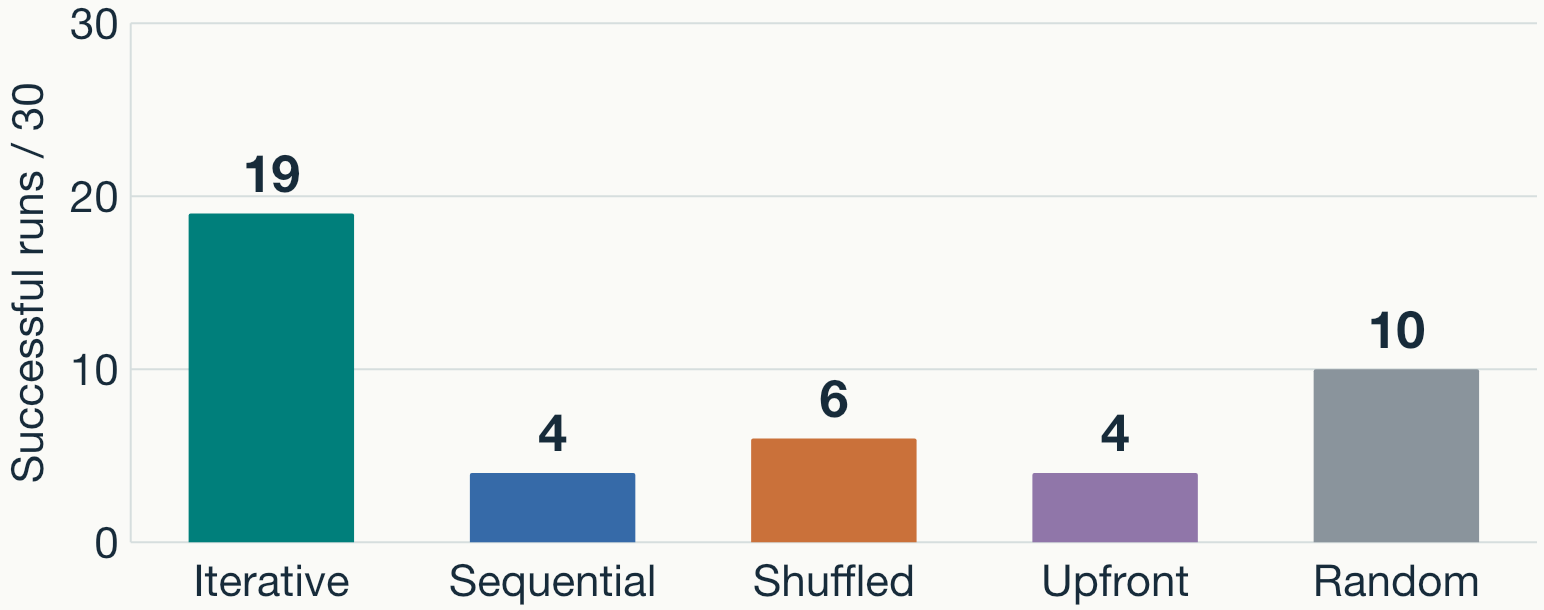
Same pool and ten charged evaluations. Thirty runs per arm.

Arm	Information available	Question answered
Iterative	Own choices and true evaluated outcomes	Can useful feedback improve selection?
Sequential	Own choices, without evaluated outcomes	What changes without result feedback?
Shuffled	Outcome values assigned to wrong candidates	Does correct attribution matter?
Upfront	One shortlist before observing outcomes	How far does prior knowledge get?
Random	Uniform sampling without replacement	Does the LLM beat broad exploration?

Haiku 5.5, temperature 1.0. Repeated or invalid queries consume budget.

The original benchmark supports feedback use

Acceptable coating found within ten queries, out of 30 runs.



19 vs 6

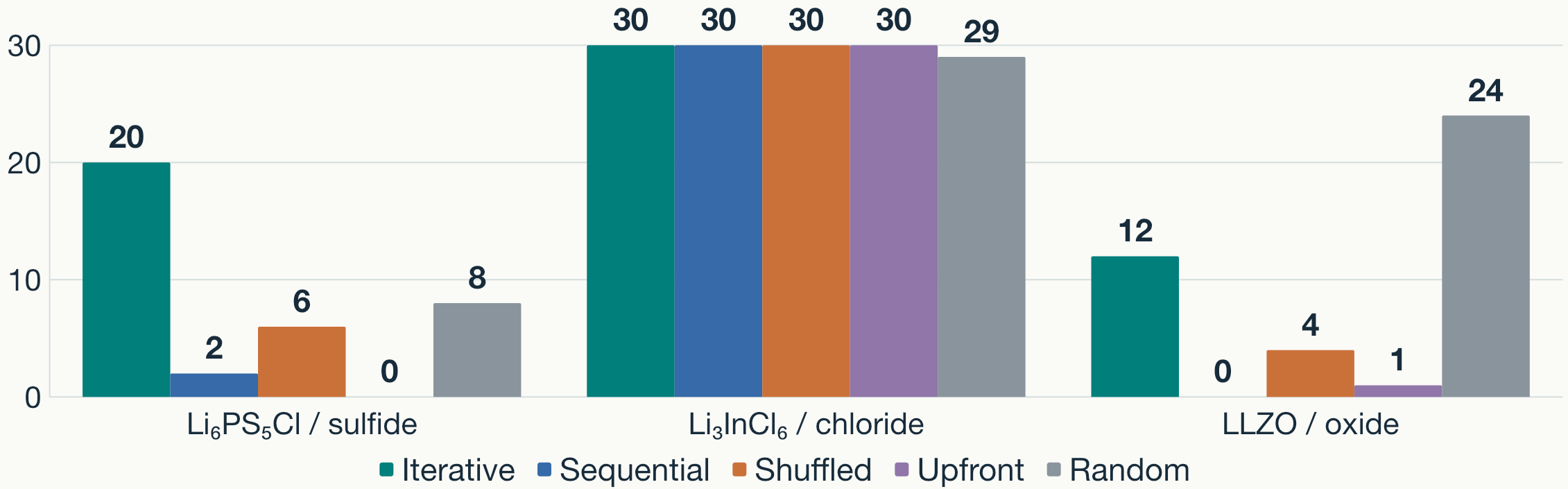
True versus shuffled feedback
Paired p = 0.0044

Random expectation
36.3% at query 10

But a post-hoc fluoride-first rule succeeds by query 2. This is not superiority over every heuristic.

Performance depends strongly on the environment

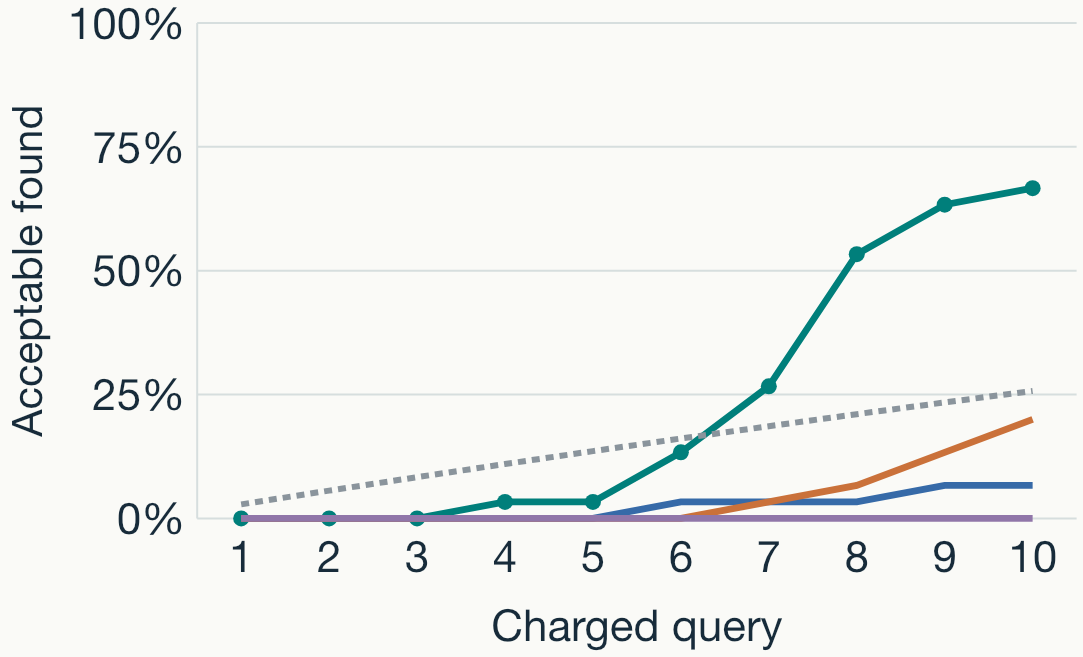
Successful runs out of 30 after ten queries. Same $LiCoO_2$ cathode.



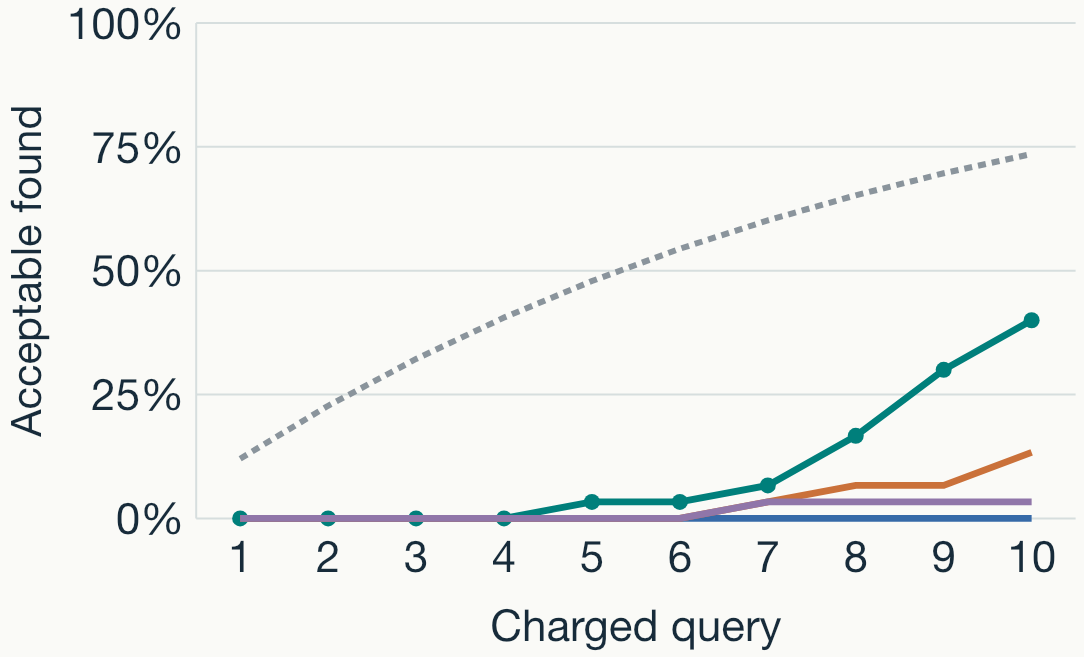
Sulfide: feedback helps. Chloride: predicted ceiling. LLZO: random wins.

On LLZO, adaptation arrives late

Sulfide



LLZO



Iterative **Sequential** **Shuffled** **Upfront** **Exact random**

LLZO: 10 of 12 iterative successes first appear at queries 8–10.

Where the LLM substitutes chemical intuition

After familiar oxides and LiF fail, Haiku refines mixed-cation fluoride chemistry.

Query	Chosen material	Observed penalty	Decision pattern
6	Li_3AlF_6	0.160	Test Al-containing fluoride
7	$LiMgAlF_6$	0.057	Change cation chemistry
8	$LiCaAlF_6$	0.048: acceptable	Refine a near-threshold candidate

The delegated act is connecting a failed result to a new chemical proposal.

Penalties: eV / non-Li atom. The proposed mechanism is a hypothesis, not a verified explanation.

A narrow prior can lose to broad exploration

Three hundred picks per arm. Random spreads its budget across more chemistry.

Arm	Distinct formulas	Acceptable picks	Successful runs
Random	95	33 / 300	24 / 30
Iterative	33	29 / 300	12 / 30
Upfront	22	1 / 300	1 / 30
Sequential	19	0 / 300	0 / 30

Iterative finds acceptable materials late and repeatedly visits $LiAlSiO_4$ polymorphs.

A post-hoc chemical audit flags 5 of 17 acceptable structures. The frozen result stays unchanged.

The expensive pilot makes intuition falsifiable

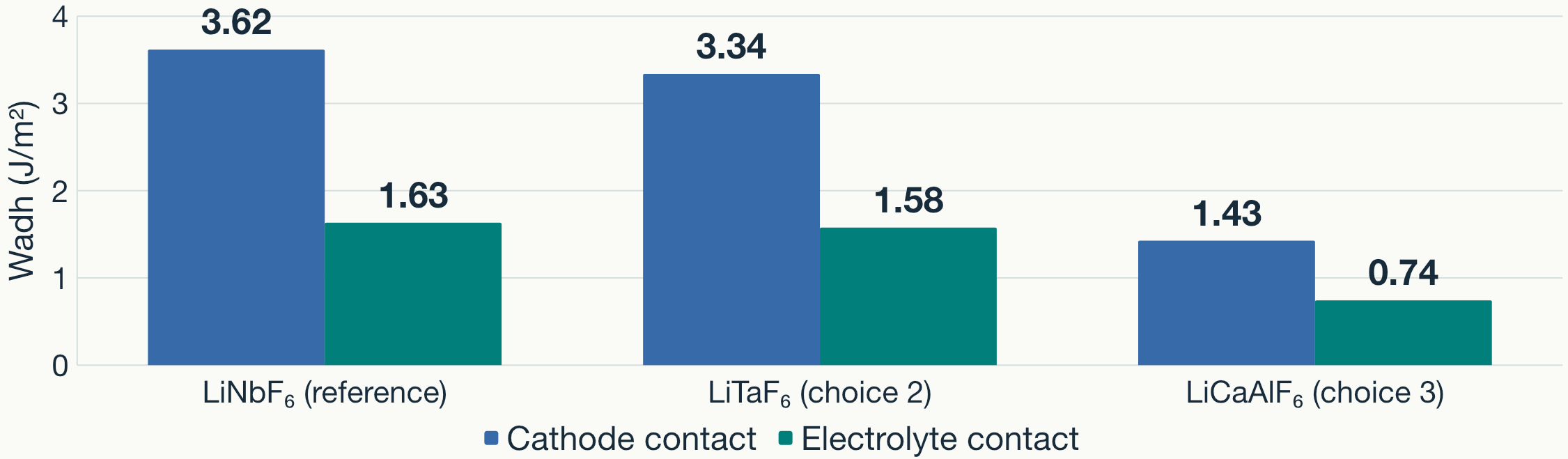
One known reference, three feasible alternatives, two new coating evaluations.

Stage	LLM responsibility	Physical evidence
Before a new coating	Choose the material and forecast both contacts	Reference $LiNbF_6$ only
After $LiTaF_6$	Assess its earlier forecasts and choose again	Six primary jobs, three registries per contact
After $LiCaAlF_6$	Explain what held and what failed	Six further primary jobs

The order stayed $LiTaF_6$ then $LiCaAlF_6$. This pilot tests predictions, not search superiority.

PET-MAD S 1.6.0 (r^2SCAN). Robustness results were released after the final LLM response.

All three coatings have a weaker electrolyte contact



Provisional adhesion for formally polar model contacts. LiF was not physically evaluated.

The analogy worked for Ta, failed for Ca–Al

All forecasts were recorded before the new coating's calculations.

Coating / contact	Forecast Wadh	Observed	Check
$LiTaF_6$ / cathode	3.2–3.9	3.339	In range
$LiTaF_6$ / electrolyte	1.4–1.9	1.577	In range
$LiCaAlF_6$ / cathode	3.0–3.8	1.427	Below range
$LiCaAlF_6$ / electrolyte	1.4–1.9	0.744	Below range

The weak contact was predicted correctly twice. Haiku acknowledged the Ca–Al forecast failure.

Units: J/m². Two material choices do not establish forecast calibration or search advantage.

A small ranking gap is not a robust materials winner

Observation	Magnitude	What we can conclude
$LiNbF_6$ versus $LiTaF_6$ weaker-contact gap	0.056 J/m²	Too small to establish a robust winner
Alternative electrolyte structure Nb and Ta	–0.121 / –0.114 J/m²	Construction sensitivity exceeds that gap
$LiCaAlF_6$ electrolyte registry range	0.523–0.744 J/m²	Substantially lower primary adhesion
$LiCaAlF_6$ electrolyte thickness and bulk structure	Unquantified	Four passing checks do not close these gaps

All contact models are formally polar. Numerical convergence is not physical validation.

What survived the original design, and what changed

Original intent	What we achieved	Assessment
Delegate scientific intuition	Choices, feedback interpretation, forecasts	Implemented and tested
Distinguish reasoning from recall	Five arms, 30 runs, shuffled feedback	Feedback benefit is conditional
Find better coatings efficiently	Exhaustive benchmark, environment transfer	No general win over random
Compare 3–5 coatings with MLIP	Three coatings, two contacts, robustness	Lower end reached; LiF unrun
Accelerate real materials discovery	Candidate prioritization under proxies	Not physically validated

Stronger evaluation than the initial plan. A narrower discovery claim than the initial ambition.

An auditable test of scientific judgment

An LLM can take over a bounded part of the scientist's job: choose a hypothesis, test it, and reassess it.

Our evidence shows where that helps: the sulfide benchmark.
It also shows where it fails: narrow exploration on LLZO
and failed transfer to $LiCaAlF_6$.

Next: improve exploration, add practical coating constraints,
and test expensive selection across more independent decisions.

Three of six planned contrasts survive correction

Environment / contrast	Discordant I : control	Raw p	Holm p
Sulfide / sequential	18 : 0	7.63e-6	4.58e-5
Sulfide / shuffled	15 : 1	5.19e-4	0.0024
Chloride / sequential	0 : 0	1.0000	1.0000
Chloride / shuffled	0 : 0	1.0000	1.0000
LLZO / sequential	12 : 0	4.88e-4	0.0024
LLZO / shuffled	11 : 3	0.0574	0.1721

Primary endpoint: at least one acceptable coating within ten charged queries.

Acceptability, optimum-finding and practicality differ

Question	Measure	Observed limitation
Can the policy find a passing candidate?	Acceptance at $P \leq 0.050$	This is the primary reported result
Does it find the pool's optimum?	Best-bin discovery / regret	Sulfide: iterative 6, random 5 LLZO: all LLMs 0, random 2 (of 30)
Is the actual coating useful?	Transport, processing, electrochemistry	Not established by this evaluator
Is the search saving compute?	Cost of avoided expensive evaluations	Cheap tables alone cannot show this

Acceptance cuts through a ranking bin. Near-threshold conclusions depend on the frozen definition.

The LLZO audit qualifies the criterion, not the score

5 of 17 acceptable structures were flagged for practical review: $LiIO_3$, $LiClO_4$ and Li_5IO_6.

Random's 24 successful runs	Count
Also found an unflagged acceptable candidate	18
Succeeded only through a flagged candidate	6

This is a descriptive partition, not a rerun with a new endpoint. Primary result: 24 versus 12.

The model is not proven to be making a safety judgment. Finite-temperature and processing limits remain.

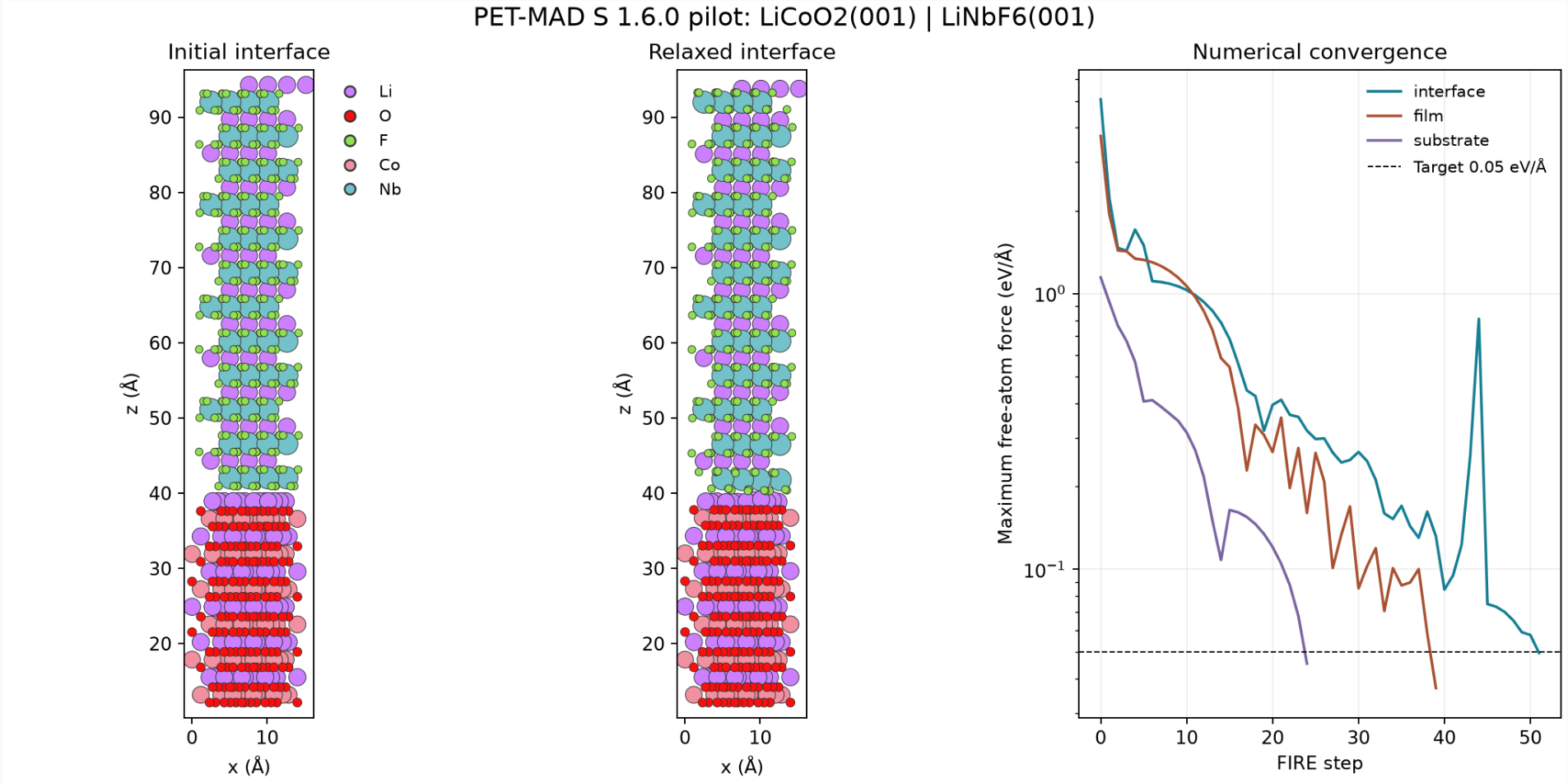
Relaxed adhesion and rigid separation are different

Coating	Cathode Wadh / Wsep	Electrolyte Wadh / Wsep	Atoms C / E
$LiNbF_6$	3.617 / 5.168	1.633 / 2.967	696 / 880
$LiTaF_6$	3.339 / 4.728	1.577 / 2.911	696 / 880
$LiCaAlF_6$	1.427 / 4.322	0.744 / 1.490	720 / 1212

Wadh: separated slabs are allowed to relax.
Wsep: fragments retain their joined-interface geometry.

J/m². Primary snapshots unchanged. Later robustness checks are a separate record.

The structures and convergence evidence are archived



The result is reproducible; integration remains separate

Evidence / cost	Verified record
Environment study	450 runs, 2,800 API calls, about $3.63 at recorded list rates
Physical pilot LLM	5 / 10 HTTP attempts, about $0.006 at recorded list rates
New PET-MAD jobs	26 / 36, all converged, 6,632 runner calls + 52 single points
Compute time	25.4 min runner time; 47.5 min summed Slurm elapsed Reused work, bulk preparation and queue time are separate
Remaining delivery work	Apply final-index reader patch, review combined diff, merge, and synchronize the structure viewer

Frozen decisions, later robustness, source hashes and limitations remain independently inspectable.